

Shiran Wang
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Education

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- **UC San Diego** **2025-present**
Bioengineering: Bioinformatics
 - Notable Courses: Java Programming & Data Structures, Differential Equations, Multivariable Calculus, Linear Algebra, Biomolecular Engineering, Mathematics for Algorithms and Systems, Genetics, Macro and Micro Economics, Mechanics, Electricity & Magnetism, Thermodynamics

Skills

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- **Technical:** Fusion 360 CAD, Arduino, Java, Rstudio Seurat, MATLAB, Mathematica, Xenium Explorer, Onshape, Meshlab, PrusaSlicer, Microsoft Office
 - **Wet Lab:** Recombinant protein expression and purification (*E. coli*), molecular cloning and PCR, agarose and polyacrylamide gel electrophoresis, affinity and size-exclusion chromatography, enzyme activity assays, tissue lysis and genotyping (mouse tail), Western blotting, mouse perfusion, paraffin sectioning
 - **Languages:** English (Bilingual), Mandarin (Bilingual)

Research Experience

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- **BMES(Biomedical Engineering Society) Project Team** **2025-present**
 - Designed and programmed a vibrotactile-feedback prosthetic arm and Haptic glove, modeled mechanical components using Fusion 360 CAD, programmed Arduino microcontrollers to control tactile actuators, integrated sensors and feedback mechanisms to enable real-time vibrotactile response.
 - Collaborated with multidisciplinary engineering teammates in design reviews and system troubleshooting.
 - **Student Research Assistant** **2026-present**
 - *University of California, San Diego*
 - Researching thrombus formation in the left atria from human induced pressure increase in mice models. Analyzed single nuclei sequencing, performed Xenium spatial transcriptomics data analysis
 - **Researcher & First Author** **2019-2024**
 - *Shanghai Institute of Immunity and Infection, Chinese Academy of Sciences*
 - Cloned, expressed, and purified recombinant human ADH1C and ALDH2 in *E. coli*, confirming enzymatic activity and cooperative detoxification of ethanol *in vitro*
 - Laid the groundwork for enzyme-based therapeutic strategies aimed at alleviating alcohol-induced toxicity
 - **Student Research Assistant** **2024-2025**
 - *University of California, San Diego*
 - Contributed to research on the therapeutic potential of SynCav1 in enhancing synaptic plasticity and neuronal survival in gene-knockout mouse models of Alzheimer's disease, advancing the development of innovative gene therapy strategies aimed at restoring neural connectivity and mitigate cognitive decline

Leadership & Community

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- **CES (Chinese Engineering Society)** **2025-present**
 - Lead of the Career Development Program, planned and coordinated professional events and networking initiatives that connected over 200 engineers across disciplines
 - **Science Olympiad** **2022-2025**
 - Led team and competed in Science Olympiad regional and state competitions. Coached Air Trajectory and Helicopter for high school and middle school teams
 - Medaled multiple times in regional competitions, placed 8th in state level competition in Engineering CAD
 - **C2K** **2022-2025**
 - Disassembled laptops and PCs, sorted electronic components for recycling, packed and distributed refurbished computers, PCs, keyboards, headsets to over 500 families in need

Extracurriculars

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- **UBIC(Undergraduate Bioinformatics Club), UCSD Sungod Archery Team, Badminton Team**

Publications

Wang, S., & Xu, Q. (2024, December 10). *Recombinant preparation and characterization of ADH1C and ALDH2 in alcohol metabolism*. Journal of Emerging Investigators. <https://emerginginvestigators.org/articles/24-135>